WORKED SOLUTION

STEP III 2003, Question 5

Pure • Cubic equations

STEP III 2003, Question 5

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/03-s3-q5

The key idea

Count roots by looking at the graph. A quadratic with positive leading coefficient crosses the x-axis twice exactly when its minimum lies below the axis. A cubic with positive leading coefficient crosses three times exactly when its maximum is above the axis and its minimum below. To make all the roots positive, look at which side of the y-axis the turning points sit, and at the sign of the curve where it meets the y-axis. The last equation is a disguised case of the shifted cubic: complete the cube to bring it to that form.

Solution

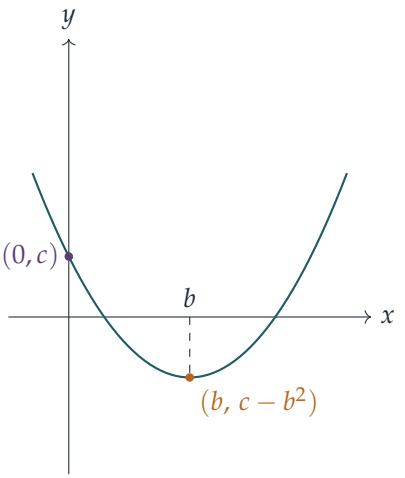
The quadratic

Completing the square,

$$y = x^2 - 2bx + c = (x-b)^2 + c - b^2,$$

so the turning point (a minimum) is at $(b,\ c-b^2)$.

When there are two distinct real roots, the curve is a $\cup$-shaped parabola, symmetric about $x = b$, whose vertex lies below the x-axis. It meets the y-axis at $(0, c)$.



Two distinct real roots. The parabola has positive leading coefficient, so $y \to +\infty$ as $x \to \pm\infty$, and its smallest value is $c - b^2$. If $c - b^2 < 0$, the curve dips below the axis and comes back up on each side of $x = b$, so it crosses twice. If $c - b^2 = 0$ it just touches the axis (a repeated root), and if $c - b^2 > 0$ it never meets it. So the condition is

$$b^2 > c.$$

Two distinct positive roots. We claim the conditions are

$$b^2 > c, \qquad b > 0, \qquad c > 0.$$

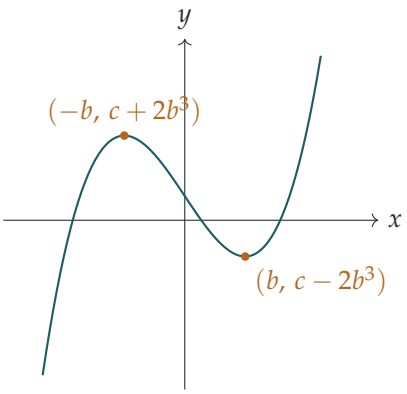
Necessary: the roots α, β satisfy $\alpha + \beta = 2b$ and $\alpha\beta = c$, and if both are positive then both of these are positive. **Sufficient:** if $b^2 > c$ there are two roots, one on each side of $x = b$. Since $b > 0$, the right-hand root is positive. The left-hand root lies in the interval where the curve is decreasing; the curve is positive at $x = 0$ (height $c > 0$) and negative at $x = b > 0$, so this root lies strictly between 0 and b, and is positive too.

The cubic

For $y = x^3 - 3b^2x + c$ we have $\dfrac{\mathrm{d}y}{\mathrm{d}x} = 3x^2 - 3b^2 = 3(x - b)(x + b)$, which is zero at $x = \pm b$. Substituting,

$$x = -b:\ y = -b^3 + 3b^3 + c = c + 2b^3, \qquad x = b:\ y = b^3 - 3b^3 + c = c - 2b^3.$$

Since $b > 0$, the turning points are a maximum at $(-b,\ c + 2b^3)$ and a minimum at $(b,\ c - 2b^3)$: the derivative is positive for $x < -b$, negative between, and positive for $x > b$.



The curve rises from $-\infty$ to the maximum, falls to the minimum, then rises to $+\infty$, and is strictly monotonic on each of these three pieces. So it has one root on each piece, giving three distinct roots, exactly when the maximum is above the axis and the minimum below:

$$c+2b^3>0 \text{ and } c-2b^3<0, \qquad \text{that is,} \qquad -2b^3<c<2b^3.$$

If either turning value is zero we get a repeated root, and if both have the same strict sign there is only one real root. So $|c|<2b^3$ is necessary and sufficient.

Three distinct positive roots of the shifted cubic

The curve $y=(x-a)^3-3b^2(x-a)+c$ is the previous curve translated a to the right (we keep $b>0$). So it has three distinct real roots if and only if $|c|<2b^3$, and then its maximum is at $x=a-b$, its minimum at $x=a+b$, and the roots $x_1<x_2<x_3$ satisfy

$$x_1<a-b<x_2<a+b<x_3.$$

All three are positive if and only if the smallest, x_1, is positive. Write

$$y(0)=(-a)^3-3b^2(-a)+c=c-a^3+3ab^2$$

for the height at which the curve crosses the y-axis. We claim that the conditions are

$$|c|<2b^3, \qquad a>b, \qquad c<a^3-3ab^2.$$

Ask yourself Why isn't "$y(0)<0$" enough on its own to put all three roots to the right of the origin?

Because y is also negative between x_2 and x_3. A negative value at $x=0$ could mean that the origin sits between the middle and largest roots. We must also know that 0 lies on the first rising piece of the curve, to the left of the maximum. That is the role of $a>b$.

Necessary. Suppose there are three distinct positive roots. Then $|c|<2b^3$ from the previous part. Also $0<x_1<a-b$, so $a>b$. Finally, y is strictly increasing on $(-\infty,a-b]$ and $y(x_1)=0$, so $y(0)<0$ because $0<x_1$. This gives $c<a^3-3ab^2$.

Sufficient. Suppose all three conditions hold. There are three distinct real roots, and $0 < a - b$, so 0 lies in $(-\infty, a-b]$, where y is strictly increasing. Since $y(0) < 0 = y(x_1)$, we get $0 < x_1$, and hence all three roots are positive.

So the equation has three distinct positive roots if and only if

$$a > b \quad \text{and} \quad -2b^3 < c < \min(2b^3,\ a^3 - 3ab^2).$$

(Since $a^3 - 3ab^2 + 2b^3 = (a-b)^2(a+2b) > 0$ when $a > b > 0$, these bounds on c are consistent.)

The final equation

Divide by 2 and complete the cube. The x^2 term in $(x-a)^3$ is $-3ax^2$, so we need $3a = \frac{9}{2}$, that is, $a = \frac{3}{2}$. Putting $x = t + \frac{3}{2}$,

$$\begin{aligned} x^3 - \tfrac{9}{2}x^2 + \tfrac{7}{2}x - \tfrac{1}{2} &= t^3 + (\tfrac{27}{4} - \tfrac{27}{2} + \tfrac{7}{2})t + (\tfrac{27}{8} - \tfrac{81}{8} + \tfrac{21}{4} - \tfrac{1}{2}) \\ &= t^3 - \tfrac{13}{4}t - 2, \end{aligned}$$

where the t^2 terms cancel by the choice of a. So

$$2x^3 - 9x^2 + 7x - 1 = 0 \iff (x - \tfrac{3}{2})^3 - 3b^2(x - \tfrac{3}{2}) + c = 0, \qquad a = \tfrac{3}{2},\ b^2 = \tfrac{13}{12},\ c = -2,$$

with $b = \sqrt{13/12} > 0$. Now we check the three conditions.

- $|c| < 2b^3$: this says $2 < 2b^3$, i.e. $b^2 > 1$, which holds because $b^2 = \frac{13}{12} > 1$.
- $a > b$: both are positive and $a^2 = \frac{9}{4} = \frac{27}{12} > \frac{13}{12} = b^2$.
- $c < a^3 - 3ab^2$: here $a^3 - 3ab^2 = a(a^2 - 3b^2) = \frac{3}{2}(\frac{9}{4} - \frac{13}{4}) = -\frac{3}{2}$, and $-2 < -\frac{3}{2}$.

All three hold, so $2x^3 - 9x^2 + 7x - 1 = 0$ has three distinct positive roots, as required.

Checks

- Sign changes confirm the final answer directly. With $p(x) = 2x^3 - 9x^2 + 7x - 1$ we have $p(0) = -1$, $p(\frac{1}{2}) = \frac{1}{2}$, $p(1) = -1$, $p(3) = -7$ and $p(4) = 11$. So there are roots in $(0, \frac{1}{2})$, $(\frac{1}{2}, 1)$ and $(3, 4)$. Numerically they are 0.185, 0.760 and 3.555.
- The value $y(0) = c - a^3 + 3ab^2 = -2 + \frac{3}{2} = -\frac{1}{2}$ matches the constant term of the monic cubic $x^3 - \frac{9}{2}x^2 + \frac{7}{2}x - \frac{1}{2}$, as it must.
- A random test of thousands of choices of (a, b, c) agreed with the stated conditions every time.

Common slips

- Using only $y(0) < 0$ (or only $a > b$) for positivity. Both are needed: $y(0) < 0$ also holds when the origin lies between the two larger roots.

- Forgetting that equality in $|c| \leq 2b^3$ gives a repeated root, so it is not "three distinct roots".
- Getting the shift the wrong way round: the curve in $x - a$ is the original curve moved a to the *right*.

Beyond STEP

Any cubic can be moved into the "depressed" form $t^3 + pt + q$ by a shift, as here. It then has three distinct real roots exactly when $4p^3 + 27q^2 < 0$; with $p = -3b^2$, $q = c$ this is the condition $c^2 < 4b^6$ found above. The quantity $-(4p^3 + 27q^2)$ is the *discriminant*, and it is the start of the story of how cubics were solved (Cardano's formula) and, eventually, of Galois theory. Allenby's *Rings, Fields and Groups* is a good next step towards that theory.

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